

Design and development of Microstrip patch antenna for vehicle to vehicle communication

Team Members:

M. Charan	24915A0416
M. Chandra Shekhar	24915A0414
A. Pratyush Reddy	23911A04D1
G. Sreethan	23911A04F1

Guide

Mrs. Sharmeen Sultana
Assistant Professor

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ABSTRACT.

Vehicle-to-Vehicle (V2V) communication systems demand compact, low-profile antennas capable of providing reliable performance in dynamic vehicular environments. This project presents the design and development of a microstrip patch antenna operating in the 5.9 GHz V2V communication band. The proposed antenna is based on a rectangular patch configuration fabricated on an RT Duroid 5880 substrate with dimensions of $36(L) \times 25(W)$ mm. The antenna geometry is optimized through full-wave electromagnetic simulation to achieve improved impedance matching and bandwidth. The compact size, simple structure, and cost-effective design make the proposed antenna suitable for integration into V2V communication modules and intelligent transportation systems. 5.9 GHz is used since regulators worldwide have specifically allocated a 75 MHz band around 5.9 GHz for intelligent transportation and vehicle-to-everything (V2X) communications, which needs short-range, high-data-rate, low-latency links for moving vehicles. . The designed antenna can provide advantages in WLAN/DSRC applications for vehicle communication

AIM & OBJECTIVES

AIM:

The main aim of this work is to design, simulate, and develop a compact microstrip patch antenna that operates efficiently in the 5.9 GHz Vehicle-to-Vehicle (V2V) communication band, providing reliable radiation performance suitable for intelligent transportation systems.

OBJECTIVES:

- To study the requirements of V2V communication systems, including operating frequency, bandwidth, gain, and radiation characteristics.
- To design a rectangular microstrip patch antenna using appropriate substrate material (RT Duriod 5880) and feeding technique suitable for vehicular applications.
- To perform electromagnetic simulation of the proposed antenna using high-frequency structure simulator (HFSS) to analyze key parameters.
- To evaluate the radiation performance of the antenna for stable and reliable V2V communication in dynamic environments.
- To develop a low-profile and cost-effective antenna design suitable for easy integration into vehicle communication systems.

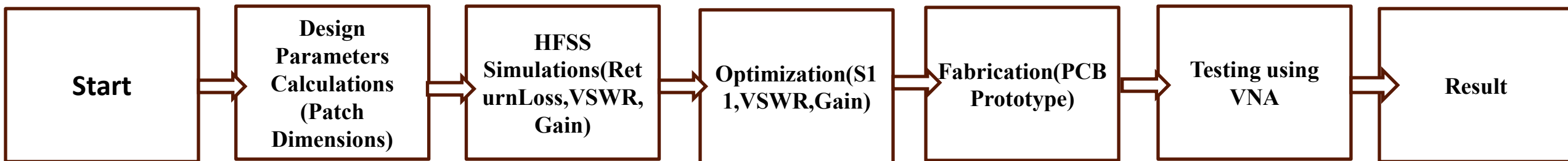
EXISTING METHODOLOGY

- Antenna Design Technique: Uses conventional microstrip patch antenna with standard geometry and minimal modification.
- Bandwidth Performance: Provides limited bandwidth(10MHz), which may not fully cover the V2V communication band.
- Gain and Radiation Efficiency: Exhibits moderate gain and efficiency, limiting communication range.
- Design and Optimization Approach: Based mainly on analytical calculations with limited simulation-based tuning.
- Vehicular Environment Suitability: Limited consideration of vehicular mounting effects and dynamic mobility conditions
- Future Standard Compatibility: Primarily designed for DSRC-based V2V communication.

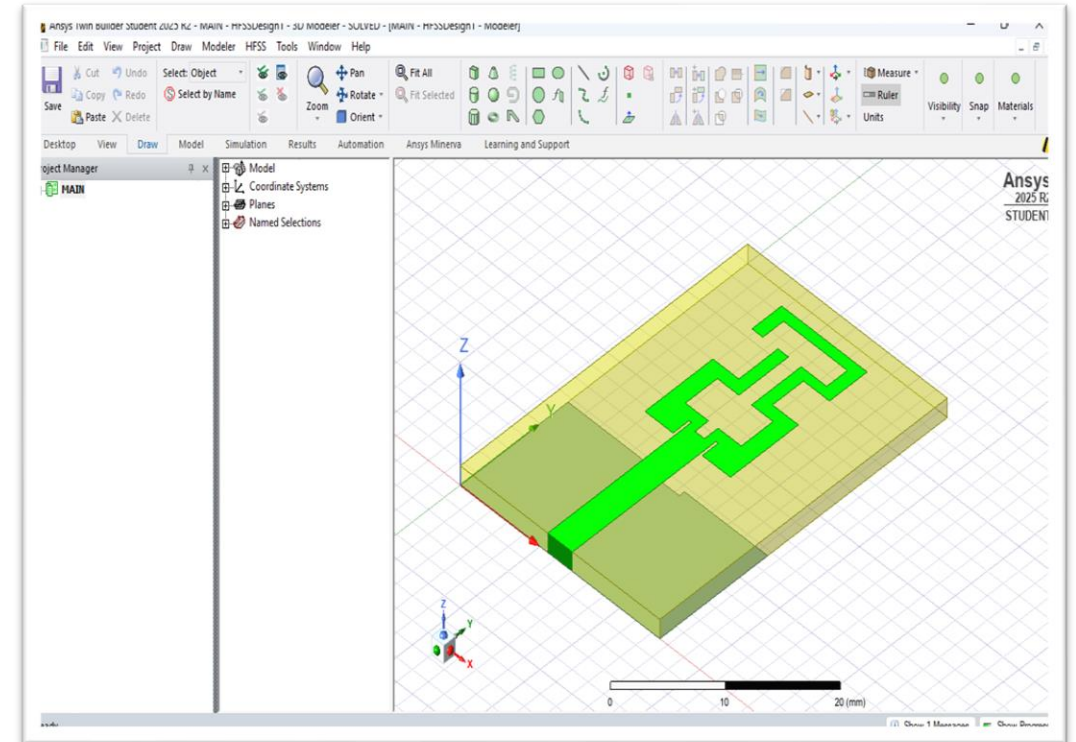
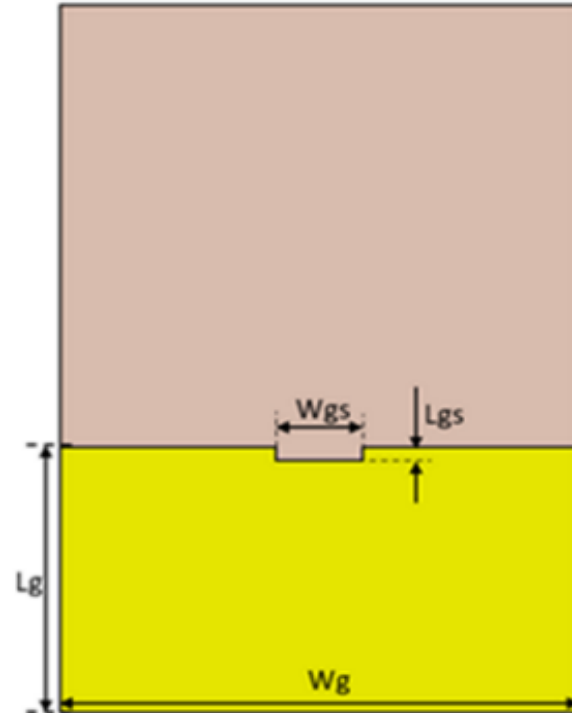
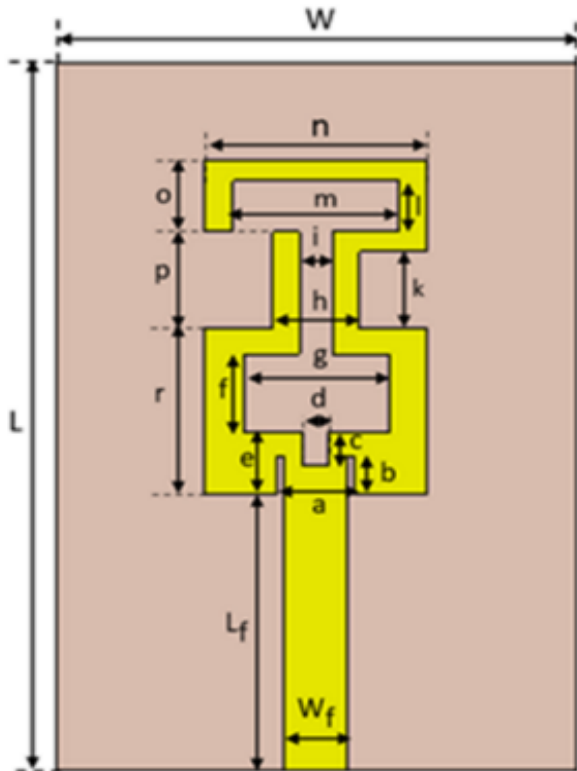
PROPOSED METHODOLOGY

- Employs an optimized microstrip patch antenna with modified geometry (slots / ground modification) to improve performance.
- Achieves enhanced bandwidth($\sim 100\text{MHz}$) through electromagnetic optimization, ensuring stable operation across the entire V2V frequency range.
- Offers improved gain and radiation efficiency, resulting in better signal strength and extended V2V communication range.
- Uses high-frequency structure simulator with parametric optimization for accurate performance enhancement.
- Designed considering vehicular environment constraints, ensuring stable radiation patterns under real driving conditions.
- Supports DSRC, C-V2X, and future 5G/6G V2V systems, making it scalable and future-ready.

BLOCK DIAGRAM

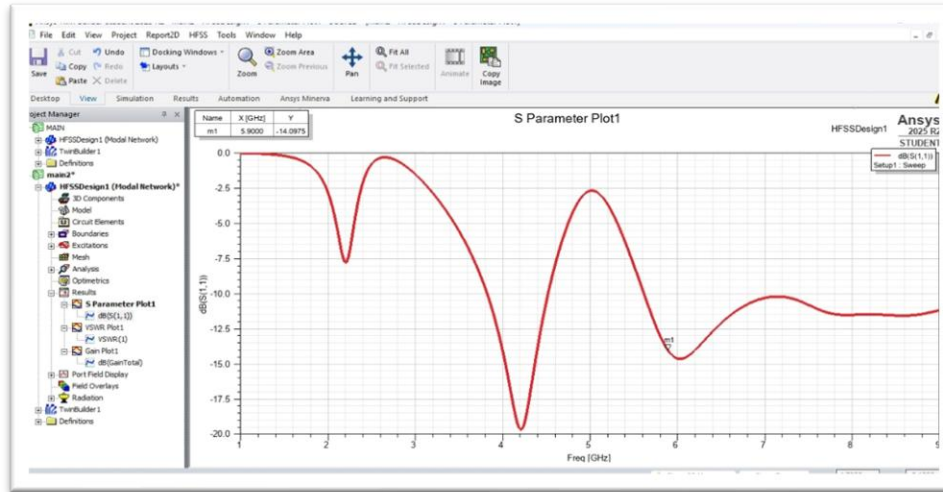


CIRCUIT DIAGRAM

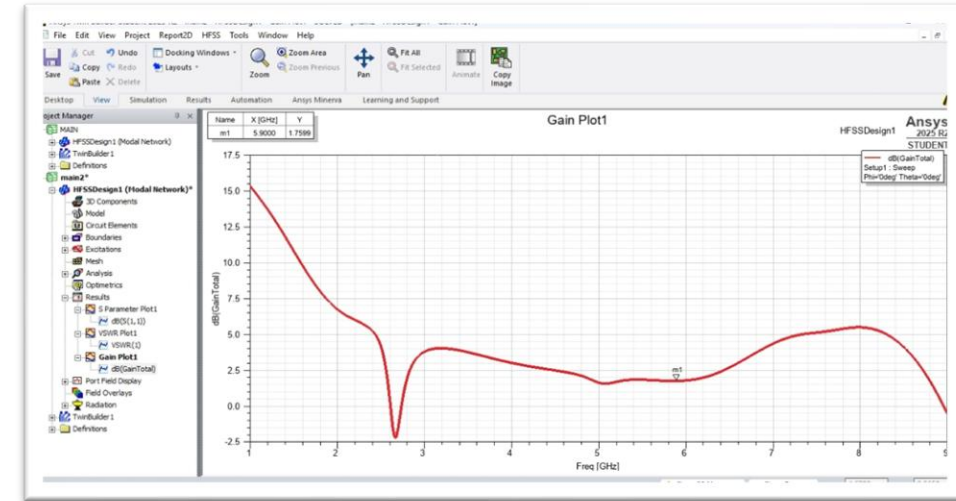


Microstrip patch antenna

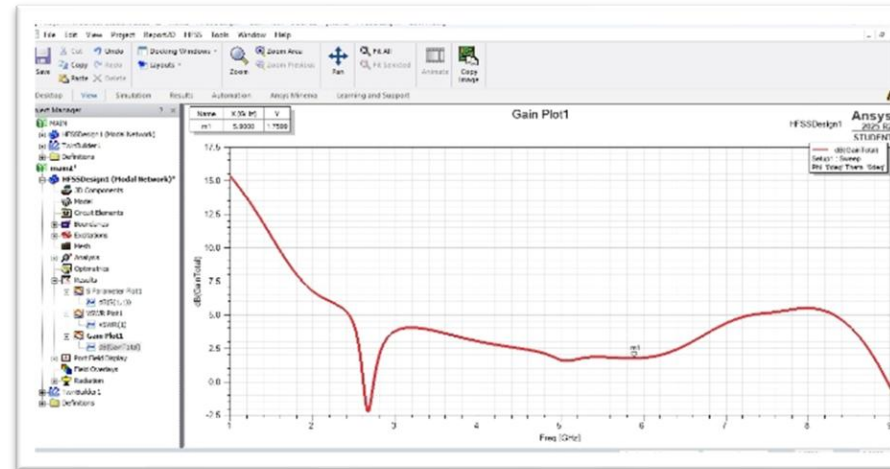
RESULTS



1.S11(return Loss)(-14dB)



2.Gain(1.75dB)



3.VSWR(1.49)

CONCLUSION AND FUTURE SCOPE

The microstrip patch antenna designed using an RT Duroid substrate demonstrates efficient performance for Vehicle-to-Vehicle (V2V) communication due to its low dielectric loss, high stability, and good radiation characteristics. Its compact size and reliable signal transmission make it suitable for integration in modern vehicular systems.

In the future, the design can be improved by enhancing bandwidth, integrating with 5G technologies, using antenna arrays or MIMO systems, and implementing beam steering techniques. Further real-time testing in dynamic vehicular environments can also help optimize performance for practical applications.

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THANK YOU